

The functional inventory of Linear A: fifty-six units with a function fixed by distribution, and what the corpus cannot fix

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Abstract

A century of proposed readings of Linear A has not produced an inventory that says, unit by unit, what is known, on what evidence, and what observation would refute it. This paper builds one for the 156 units of the corpus with three or more intact attestations. Each reading is fixed by the unit's distribution in the corpus (position, figures and fractions, company, site and seal) against a permutation null with a measured error rate, and is filed with the evidence for, the evidence against, and a refutation condition. The result is 56 units with a function: four arithmetic terms, eight headings and transaction terms, sixteen signs of commodities, livestock and persons, sixteen qualifiers and grades, eight place and personal names shared with Linear B, and four isolated functions. Fourteen have no prior source; the other 42 have prior art and carry a measurement here for the first time. None is a word with lexical meaning: the logograms are pictures, and the syllabic sign-groups have a function, not a translation. The inventory also brings three facts about the corpus that the editions do not state: 276 of the 783 syllabic units are broken in every attestation (35%), which invalidates part of the published morphological alternations; the 801 sealed nodules of Haghia Triada form two seal circuits that share no signs, and the sign does not identify the sealer; and the archive records no metal, honey or fish, all present in the material culture. The ceiling is declared: with this corpus, the internal method fixes functions and not words, and the number moves not with more runs but with new data.

Keywords: Linear A, Haghia Triada, inventory, null models, logograms, qualifiers, nodules, error rate.

1. Why an inventory and not one more reading

Whoever looks in Linear A for what is known about a particular unit finds Younger's lexicon, which collects almost everything and does not distinguish the measured from the conjectured, and a bibliography of proposals that seldom states what datum would refute them. What does not exist is a table with three columns: what is claimed, what supports it, what would bring it down. This paper is that table, built with one procedure and published with its code.

The procedure does not read. It fixes what a unit **does** in the account from how the scribe treats it: whether it opens the document, whether it carries a figure, whether that figure admits fractions, whether it follows a commodity with a smaller figure, whether a syllable is added to it or it is the added syllable, which other units share its document, and on nodules, which seal. Each of those observations has a null that preserves what the claim does not explain, and each instrument has an error rate measured on material where it should find nothing.

2. The corpus, and what the editions do not say about it

The corpus is GORILA through the LinearA Explorer: 1,720 documents and 1,007 distinct units, of which 156 have three or more attestations and at least one intact. That is the ceiling of the inventory: below three attestations there is no profile to measure.

The broken units. The words field of the edition keeps the break sign and the transliteration drops it. Recovered, 276 of the 783 syllabic units (35%) are broken in every attestation:]MA-TE-RE is not MA-TE-RE. Three of the eight lexical coincidences with Linear B of a first comparison were fragments, and of Davis's stem/ending pairs, five of eight endings fall below three stems when only intact units count (-JA, -ME and -TI remain). The whole inventory is computed on intact units.

The nodules. Younger's commentaries give, for 801 nodules of Haghia Triada, the inscribed sign and the seal (CMS II,6 number and motif). Sign and seal are associated (mutual information 0.88 against 0.19 in 500 permutations), but not one to one: the seals form two blocks that share no signs (AT 13, 99, 45, 9, 17, 38 only with 301 and ZE; AT 125, 105, 19, 95, 79 with KU, KA, SI, RO, I and TA and never 301), and within a block one seal uses six signs. The sign does not identify the sealer; it identifies what is sealed. AT 19 uses KU on 42 of 44 nodules: one official for one type. What is missing to read those signs is the nodule type per document (Hallager 1996, vol. II), which is not in any open source.

The gaps. The material culture of Haghia Triada has copper ingots, honey and fish. No sign with a figure has the profile of those things (low whole numbers or weight for metal, liquid fractions for honey), *118 is the weight unit and accompanies cyperus and sesame, and Younger's lexicon does not mention them. The tablet archive does not record them.

3. The instruments and their error rates

Twelve instruments, each with its null and its measured rate (data/derived/error_rates.json). Those that support the inventory:

instrument	what it measures	null	error rate at 0.05
quantity profile	fraction rate of a logogram's figures against the base (27%), in the direction the class predicts; size of the figures	fractions at the base rate	4.6% (worst case, direction chosen a posteriori)
heading	first position on a tablet against the base (11.8%)	binomial	positional: 3-5%
sub-count	figure smaller than	binomial	(within positional)

instrument	what it measures	null	error rate at 0.05
	the preceding commodity's against the base (52%)		
qualifier	a syllable added to two or more distinct commodity signs	a rule, not a test	—
stratified co-occurrence	two units in the same document, preserving site and margins	bipartite by site	5.5%
adjacency	two units contiguous, over eligible pairs	permutation	2.5%
name matching	exact coincidence with the Linear B lexicon by length	syllable bigrams of the corpus itself	(by length; 2 syllables = chance)
arithmetic	the figure after the unit equals the sum of those before	permutation	2.5%

An instrument is calibrated on the material it is applied to: the profile rate was measured by shuffling the fractions of already-read logograms at the base rate; the co-occurrence rate, on words and logograms of the corpus itself (8.0% unstratified, 5.5% stratified by site: the site was the confounder).

4. The inventory

Fifty-six units: 25 established (field consensus confirmed by our test), 27 proposed (our test without prior consensus, or prior art without a test of its own until now), 4 disputed (a source against). Status, attestations, reading, prior source for, and number of pieces of evidence against:

unit	status	attest.	reading	prior source	against
Arithmetic					
KU-RO	established	37	total	Godart & Olivier / field consensus, Scho	
KI-RO	established	16	deficit	Schoep, Uchitel	
DA-I	proposed	2	a total, of a	—	

unit	status	attest.	reading	prior source	against
Qualifiers and grades			kind distinct from KU-RO		
	PO-TO-KU-RO	established	2	grand total	Schoep
	KU	established	170	as an added syllable, a qualifier of four commodities (GRA+KU, TELA+KU, *188+KU,	—
	KA	proposed	169	as an added syllable, a qualifier of wine and of persons (VIN+KA, VIR+KA); alone	—
	OLE+MI	proposed	19	a grade of oil; listed together with OLE+DI in every document where it carries a	Younger 1
KI	established	19	as an added syllable, a qualifier of oil and 316 (OLE+KI, 316+KI) and	Del Freo, via Nosch and Weilhartner	

unit	status	attest.	reading	prior source	against
			of wool		
CYP+D	proposed	19	a grade of cyperus recorded in the smallest amounts (fractions in 13 of 16, maxi	—	
PA	proposed	16	qualifier (a grade or type) shared by grain, cyperus and <i>304:</i> <i>GRA+PA</i> , <i>CYP+PA</i> ,	—	
OLE+DI	proposed	12	a grade of oil; listed together with OLE+MI in every document where it carries a	Younger	1
TU	established	11	qualifier (a grade or type) restricted to the olive family: OLE+TU and OLIV+TU	Younger	
RA	proposed	9	as an added syllable, a qualifier of oil and wine	—	

unit	status	attest.	reading	prior source	against
			(OLE+RA, VIN+RA)		
OLE+TA	proposed	8	a grade of oil recorded in small fractional amounts	Younger	
GRA+KU	proposed	7	GRA with the qualifier KU (a grade or type)	—	
VIN+RA	proposed	4	wine with the qualifier RA (a grade or type), parallel to OLE+RA	—	
MA-RU-ME	proposed	3	wool with the qualifier ME (a type or grade)	Del Freo, via Nosch and Weilhartner	
*188+KU	proposed	2	*188 with the qualifier KU (a grade or type)	—	
KI-MA-RU	proposed	1	wool with the qualifier KI	Del Freo, via Nosch and Weilhartner	
TELA+KU	proposed	1	TELA with the qualifier KU (a grade or type)	—	
Headings and transaction					
TE	established	58	transaction	Schoep,	

unit	status	attest.	reading	prior source	against
			term set between a party and the commodity ("X · TE · VIN"); also th	Younger	
SA-RA ₂	disputed	20	transaction term, Haghia Triada only; possibly the ethnic of the site	Schoep, Uchitel	1
A-DU	established	10	heading term (record label) of HT tablets; heads the lists of HT 86 and HT 95 am	Younger	
KA-PA	established	6	heading term of HT tablets (HT only); KA-PA-QE is its extended form (Salgarella	Salgarella	
JE-DI	proposed	4	heading term (first position in 3 of 4 tablet attestations)	—	
A-KA-RU	established	3	heading term of HT tablets (HT 2.1, HT 86a.1, HT	Younger	

unit	status	attest.	reading	prior source	against
DA-QE-RA	proposed	3	86b.1 as second heading)	Younger	
			heading term of HT tablets (HT 6a.6 as a second heading, HT 120.1)		
KI-RI-TA ₂	proposed	2	heading term (first position in both attestations)	—	
Commodities, livestock and persons					
NI	established	76	figs (the logogram NI; ni- as the first syllable of the Minoan word)	Neumann; Salgarella and Petrakis (eds.),	
GRA	established	62	grain (cereal), the bulk dry commodity	Ventris and Chadwick	
VIN	established	53	wine, a liquid commodity	Ventris and Chadwick	
CYP	established	52	cyperus (*303), a fine commodity measured in small fractional amounts	Ventris and Chadwick	
OLIV	established	24	olives, the	Ventris and	

unit	status	attest.	reading	prior source	against
			fruit as distinct from its oil	Chadwick	
OLE+U	proposed	22	a type of oil counted in whole units (or a counted unit of oil: whole numbers, s	—	1
OLE	established	22	olive oil, a liquid commodity	Ventris and Chadwick	
*22F	established	14	goat, female; Linear B CAPf	GORILA / Ventris and Chadwick	
*308	proposed	12	fatty product derived from the olive	Koh & Birney, Younger	
*305	proposed	12	a category of persons	Younger	
*21M	established	8	sheep, male (ram); Linear B OVISm	GORILA / Ventris and Chadwick	
*23M	established	7	ox/bull; Linear B BOSm	GORILA / Ventris and Chadwick	
*21F	established	4	sheep, female (ewe); Linear B OVISf	GORILA / Ventris and Chadwick	
*131B	proposed	4	a variant of the wine sign (AB 131b), a type of wine	GORILA	
*22M	established	3	goat, male;	GORILA /	

unit	status	attest.	reading	prior source	against
			Linear B CAPm	Ventris and Chadwick	
VIR	established	0	person (man), the unit counted in personnel lists	Ventris and Chadwick	
Other functions					
JA-SA-SA- RA-ME	proposed	7	does not designate the offering or the object	—	
*306	disputed	7	possibly the sign for woman	Uchitel	2
KU-NI-SU	disputed	5	first name in a list; the Semitic reading as emmer is proposed and contested	Younger, various, cited by Younger	1
SU-KI-RI- TE-I-JA	proposed	1	formed on the stem of Sybrita, or sharing its lexicon	Davis	
Place and personal names					
KI-DA-RO	proposed	2	a personal name shared with the Linear B tablets of	Godart	

unit	status	attest.	reading	prior source	against
			Knossos (Godart 1984)		
PA-I-TO	established	2	Phaistos, the place name (Linear B pa- i-to)	Godart	
DA-I-PI-TA	proposed	2	a personal name shared with the Linear B tablets of Knossos (Godart 1984)	Godart	
I-TA-JA	proposed	1	a personal name shared with the Linear B tablets of Knossos (Godart 1984)	Godart	
DA-MA-TE	disputed	1	coincides with Linear B da-ma-te; read as Demeter by some, contested	Owens	2
SU-KI-RI- TA	established	1	Sybrita, the place name (Linear B su- ki-ri-ta)	Godart	
I-JA-TE	proposed	1	coincides with Linear B i-ja-te (PY	Godart, Younger	

unit	status	attest.	reading	prior source	against
			Eq 146); Younger: “of/from I- JA”		
SE-TO-I-JA	established	1	Setoia, the place name (Linear B se- to-i-ja; Archanes or Iouktas?)	Godart	

Four things are visible in the whole table.

Fourteen with no prior source. DA-I as a total of another kind; KI-RI-TA₂ and JE-DI as headings; VIN+RA parallel to OLE+RA; KU as a qualifier of four commodities and PA of three; KA of wine and of persons; CYP+D as cyperus in the smallest amounts; OLE+TA as fine oil; OLE+DI and OLE+MI as paired grades; JA-SA-SA-RA-ME as what it is not. They are functions, and several are the same finding seen in several units (KU in four ligatures).

Forty-two with prior art, where the measurement is what is new. VIR in whole numbers only (2 of 36 fractional, $p = 0.0008$); GRA in large whole numbers (12 of 90, maximum 976); CYP almost always in fractions and in small amounts (39 of 64, maximum 23); TE in the pattern separator-TE-separator-commodity in 10 of 24 against 2.5% ($p = 10^{-10}$); the sex-marked livestock in whole numbers (2 of 28, $p = 0.007$) with PH 31 as a list by species and sex; A-DU in first position in 7 of 10 ($p < 0.0001$). And in the other direction: six of Younger’s “headings” with zero first positions, KU-NI-SU as emmer disputed, and “*304 always between OLE and OLIV” false as stated (6 of 25).

No word. All 56 are function: what the unit does in the account. For the 16 signs of commodities, livestock and persons, the reading comes from pictographic identity with Linear B (OLE is oil because the sign is the same and in Linear B it is fixed by Greek context), and our test says the administration counted them as that kind of thing is counted. No syllabic group has a translation: KU-RO is “total” because it adds up, not because we know how “total” was said in Minoan.

What was tested and not fixed. List entries (the most numerous class) are judged only by exclusion: no fractions of their own, no added syllable, not a heading, not a total; and that exclusion leaves a personal name and a commodity written in syllables indistinguishable. Without an external source there is no positive judge, and the category “list entry” does not count as a reading. The nodule signs (*301, KU, KA, SI, RO, ZE) have their structure fixed (the two circuits) and not their reading. The five borderline units of two runs (VIR+[?], QA2+[?]+PU,

SA-RU, A-SE, JE-DI in part) are not filed: bare p between 0.01 and 0.04 with 50-80 tests is what chance produces.

4 bis. The scope of the inventory: which archive fixed each function

The 56 entries were not fixed on “Linear A” but on the archives with enough material, which is almost always Haghia Triada (64% of the corpus). The distinction is not formal. The second archive, Khania (227 documents, 105 tablets), **uses no summation term on any of its tablets**, against 41 of 205 at Haghia Triada: under the Haghia Triada rate, the probability of that zero is $7 \cdot 10^{-11}$. KU-RO, KI-RO, PO-TO-KU-RO and DA-I are functions of the Haghia Triada archive, and Khania offers nothing against which to check them.

Nor is Khania the same kind of archive: its tablets carry half the units (7.4 against 13.8) and a quarter of the syllabic groups (0.8 against 3.1), its figures are smaller (median 3 against 5), 36% of them are fractional against 17%, and its dominant commodity is cyperus (65 occurrences against 14), with 68% of its records in fractions. It is fine-grained distribution of a valued product, not bulk storage.

With the instruments run on the archive’s own base rates, Khania yields no headings and no qualifiers of its own, and yields one reading that Haghia Triada cannot give: **CYP+D carries the quantity $\frac{1}{2}$ in 12 of its 16 quantified attestations** against a 22% base ($p = 10^{-5}$), and it is issued after entries of persons (KH 7a: VIR+313b 10 then CYP+D $\frac{1}{2}$; VIR+313b 4 then CYP+D $\frac{1}{3}$). No commodity at Haghia Triada shows a dominant quantity at that level. It is the corpus’s only candidate for a fixed-quantity allocation.

The inventory is therefore to be read with this restriction: **it is the functional inventory of the Haghia Triada archive, with additions from Khania and from the votive vessels**. The Linear A corpus does not have one accounting format but at least two.

5. How it was reached, and why the number stops

The inventory went from 18 to 56 in two days of work with the same procedure on the same units: the parallel with Linear B (livestock, commodities, NI, TE), the extrapolation of structures (qualifier, sub-count, heading), name matching with a null, and two stages with a language model in the loop. Those two stages were measured: reading passages of the literature, the model invents non-existent units in 22% of cases; proposing a reading from the dossier of each unit (figures, position, company, seal), in 1%. Its contribution was routing at scale, not the idea: the three new readings it brought came from mechanical rules applied to units nobody had looked at with them. A second pass of the same procedure over the remaining 117 units found nothing.

That is the ceiling, and it should be stated precisely. It is not the ceiling of the problem: it is the ceiling of what can be **verified** with this corpus. Accounts have few functions and almost all are fixed; those that remain (some twenty nodule signs) wait for an external datum; and the step from function to word requires a judge the corpus does not contain, because nothing in it can say that “KU-NI-SU means emmer” is false. That judge would be given by a bilingual or by new running text, and those come from excavation.

6. Discussion

What this inventory changes is not the number of readings but their status. Each of the 56 carries in writing what observation would refute it, and the dictionary refuses to file an entry without that line. Each instrument carries its error rate, and a rate that came out wrong (the uniform null for fractions, 16.5%) is kept as a record of why the null was changed and not the instrument. Each claim with prior art carries the citation and the measurement together, and those that fall are filed as fallen, with the source against. It is what the field asks of a decipherment and has not asked of a lexicon: that one can know, of each line, how much it is worth.

And what the inventory says about Linear A, read whole, is something a century of proposals had not said in this form: **we know what was counted and how the account was organised, we know three places, and we know no word**. The distance between an archive understood and a language read is exactly that, and no internal method closes it.

Data and reproducibility

GORILA corpus via the LinearA Explorer (R. Hogan); nodule seals from the commentaries of J. Younger; Linear B from linearb.xyz. Dictionary, error rates, sign × seal table and runs in data/derived/ of the kuro package (github.com/BiomeMakers/kuro), with the scripts that rebuild every non-redistributable source.

Declaration of assistance

The corpus processing, the tests and the drafting of this text were produced with the assistance of a language model (Claude, Anthropic) under the direction of the author, who is responsible for all claims made here. Two stages of the procedure use the model in the loop; their invention rates are declared in section 5.

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